WORKED SOLUTION

STEP I 2016, Question 7

Pure • Factorisation Modulo Four

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q7

This question has a full guided treatment in *Beyond STEP*, Section 1.7, *When Factorisation Fails*: https://beyondstep.keithmcnulty.org/ch01/factorisation-fails.html

The key idea

Everything follows from working modulo 4. Odd numbers are $\equiv 1$ or $\equiv 3 \equiv -1 \pmod 4$, so a product of odd numbers lies in S or T according to whether it has an even or odd number of factors from T. That one parity rule does parts (ii), (iii) and (iv)(a). Part (iv)(b) asks you to see that S-primes need not be primes, and that loses unique factorisation.

Solution

Part (i)

Every odd positive integer leaves remainder 1 or 3 on division by 4, and no even number does. So $S \cup T$ is the set of all *odd positive integers*. No integer leaves two different remainders, so $S \cap T$ is the *empty set*.

Part (ii)

Take $4a+1$ and $4b+1$ in S, with $a, b \geq 0$ integers. Then

$$(4a+1)(4b+1) = 16ab + 4a + 4b + 1 = 4(4ab + a + b) + 1,$$

which is a positive integer leaving remainder 1, so it is in S.

For T, the answer is *no*. For example $3 \in T$ but $3 \times 3 = 9 = 4 \times 2 + 1 \in S$, and $S \cap T$ is empty. In fact this always happens:

$$(4a+3)(4b+3) = 16ab + 12a + 12b + 9 = 4(4ab + 3a + 3b + 2) + 1 \in S.$$

We will also need the mixed product:

$$(4a+1)(4b+3) = 16ab + 12a + 4b + 3 = 4(4ab + 3a + b) + 3 \in T.$$

In short, writing T-numbers as $\equiv -1 \pmod 4$: a product of odd numbers is $\equiv (-1)^k \pmod 4$, where k is the number of factors from T. So the product is in S if k is even and in T if k is odd.

Part (iii)

Let $n \in T$ be composite, and write $n = p_1 p_2 \cdots p_r$ as a product of primes. Since n is odd, every p_i is odd, so by part (i) each p_i is in S or in T.

Suppose, for contradiction, that every p_i is in S. Applying part (ii) repeatedly, $p_1 p_2 \in S$, then $(p_1 p_2)p_3 \in S$, and so on, so $n \in S$. But $n \in T$ and $S \cap T = \varnothing$, which is a contradiction. Hence at least one prime factor of n lies in T. (In parity terms: $n \equiv -1$ needs an odd number, so at least one, of factors $\equiv -1$.)

Part (iv)(a)

Note first that $1 \notin T$, so every element of T is at least 3.

Every $n \in T$ is T-prime or a product of T-primes. We use strong induction on n. If n is T-prime, we are done. Otherwise $n = t_1 t_2 \cdots t_k$ with $k \geq 2$ and each $t_i \in T$. Each $t_i \geq 3$, so each $t_i \leq n/3 < n$, and by the induction hypothesis each t_i is T-prime or a product of T-primes. Substituting these into $n = t_1 \cdots t_k$ writes n as a product of T-primes.

> **Ask yourself** We have a product of T-primes, but why must there be an *odd* number of them?

Every T-prime is in T, so it is $\equiv -1 \pmod 4$. A product of m of them is therefore $\equiv (-1)^m \pmod 4$, by the rule at the end of part (ii). If m were even, the product would be $\equiv 1$, so it would lie in S, not T. Since the product is $n \in T$, m is odd. So every $n \in T$ is either T-prime or the product of an odd number of T-primes, as required.

Note that T-primes need not be primes: $15 = 3 \times 5$ is T-prime, because $5 \notin T$ and the only factorisation into numbers bigger than 1 is 3×5.

Part (iv)(b)

The same idea as the remark about 15 works in S: a product of two primes from T lies in S but has no factorisation within S. Take

$$9 = 3 \times 3, \qquad 21 = 3 \times 7, \qquad 49 = 7 \times 7.$$

Each is in S. Each is S-prime: its only divisors other than 1 and itself are 3 or 7, which are in T, not S. (For 21, the divisors are $1, 3, 7, 21$.) Now

$$441 = 9 \times 49 = 21 \times 21,$$

and $441 = 4 \times 110 + 1 \in S$. These are two genuinely different factorisations of 441 into S-primes: the first uses 9 and 49, the second uses 21 twice. So 441 is an example.

? **Ask yourself** How would you have found this?

In the ordinary integers, $441 = 3^2 \times 7^2$, and the primes $3, 7$ are in T. Inside S we cannot split off a single 3 or 7, so the T-primes must be paired up. With two 3s and two 7s there are two ways to pair them: $(3 \cdot 3)(7 \cdot 7)$ or $(3 \cdot 7)(3 \cdot 7)$. Any p^2q^2 with distinct primes $p, q \in T$ works in the same way, such as $9 \times 121 = 33 \times 33$.

Checks

- Try small cases: $5 \times 9 = 45 \in S$, $3 \times 7 = 21 \in S$, $5 \times 7 = 35 \in T$.
- For (iii): $15 = 3 \times 5$ and $35 = 5 \times 7$ are in T, and each has a prime factor (3, 7) in T.
- By computer, for every $n \in T$ below 3000, every factorisation into T-primes has an odd number of factors. The T-primes start $3, 7, 11, 15, 19, 23, 31, 35, \ldots$. Note $27 = 3 \times 3 \times 3$ is not T-prime: three factors, which is odd.

Common slips

- Claiming that the product of two members of T is in T without testing a case: $3 \times 3 = 9$ settles it at once.
- Forgetting that an X-prime need not be a prime. This is the whole point of (iv)(b): 9 and 21 are S-prime.
- In (iv)(a), only proving that the number of factors is odd and forgetting to show that a factorisation into T-primes exists at all.

Beyond STEP

The set S is closed under multiplication, and its "primes" do not factorise uniquely. This is a classic toy model (sometimes called the Hilbert numbers) of what goes wrong in rings such as $\mathbb{Z}[\sqrt{-5}]$, where $6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$. Repairing unique factorisation in such rings led to ideals and algebraic number theory. Davenport's *The Higher Arithmetic* explains why unique factorisation in $\mathbb{Z}$ is a theorem that needs proof, and Allenby's *Rings, Fields and Groups* covers rings where it fails.

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